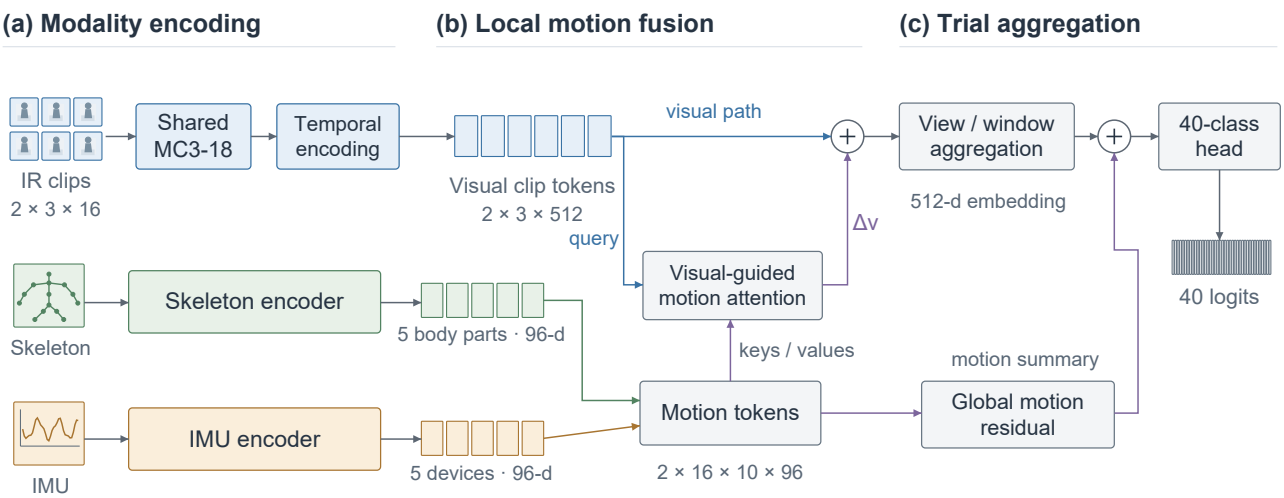

(a) Modality encoding
(b) Local motion fusion
(c) Trial aggregation
IR clips
2 × 3 × 16
Shared MC3-18
Temporal encoding
Visual clip tokens
2 × 3 × 512
visual path
query
Δv
View / window aggregation
512-d embedding
40-class head
40 logits
Skeleton
Skeleton encoder
5 body parts · 96-d
Visual-guided motion attention
keys / values
motion summary
IMU
IMU encoder
5 devices · 96-d
Motion tokens
2 × 16 × 10 × 96
Global motion residual
Local residual before view/window aggregation; global residual after aggregation.